Taxonomic DATABASE – Hyolitha

Complete Build Guide

Step-by-step documentation for all 11 code components and patches

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00 Overview

Prerequisites – Building process

Overview

This repository contains all code necessary to build and operate a relational taxonomic database for Hyolitha (extinct marine invertebrates). The system was developed at the National Museum, Czech Republic as part of an AI-assisted data extraction pipeline.

Complete Build Sequence — All 16 Steps

01–05 · Data preparation and database creation (Python, VBScript, SQL)
06–11 · Core VBA packages — main form, compare, import/export, utilities, preview, detail editor
12 · BALIK_6_7 — filter presets, scope-aware export, bulk edit, frmMergePreview, frmValidator fix
13 · BALIK_8 — frmTextEditor for Compare, CROSS-SECTION / CARDINAL_PROCESSES fix in Compare
14 · BALIK_9 — BulkRename preview crash fix, hardcoded column index map for Compare fields
15 · BALIK_10 — IMPORT XLSX / BIBTEX / DIFF VIEWER buttons, Ctrl+B shortcut, report overflow fix
16 · BALIK_11 — NOTES field in TAXONOMY tab, left sidebar navigation, CARDINAL filter-clear fix

Steps 12–16 (shown in orange) are mandatory patch packages applied after the core build. The Complete Build Workflow table in §3 shows the full sequence with one-line descriptions of each step.

The build process follows a strict sequential order across three technology layers

Google Colab (Python/pandas) — data preparation and Excel export
Windows Script Host (VBScript WSF) — database creation and import
Microsoft Access VBA — form builder packages (6 packages)

Step	File	Technology	Purpose
01	Colab_create_vw_Complete_Taxa.txt	Python / Google Colab	Prepare structured Excel with 6 thematic sheets
02	code_create_database.wsf	VBScript / WSH	Create Access DB and import all data from Excel
03	code_access_table.txt	SQL (Access)	Master query joining all 6 tables
04	create_column_compare.txt	SQL (Access)	Add Compare Yes/No flag column to Taxa
05	code_create_table_history.txt	SQL (Access)	Create audit/history table for undo support
06	code_main_form.txt	VBA (Access)	Package 1 — main search form frmDark + subform
07	code_compare_merge.txt	VBA (Access)	Package 2 — side-by-side taxon comparison form
08	code_export_import.txt	VBA (Access)	Package 3 — CSV/XLSX import, DOCX/BibTeX export
09	code_utilities.txt	VBA (Access)	Package 4 — Find/Replace, Validator, History, Stats
10	code_preview.txt	VBA (Access)	Package 5 — Quick Preview panel + printable report
11	code_detail.txt	VBA (Access)	Detail form frmDetailTaxa — full tabbed record editor
12	BALIK_6-7.bas	VBA (Access)	filter presets, scope-aware export, bulk edit, frmMergePreview, frmValidator fix
13	BALIK_8.bas	VBA (Access)	frmTextEditor for Compare, CROSS-SECTION / CARDINAL_PROCESSES fix in Compare
14	BALIK_9.bas	VBA (Access)	BulkRename preview crash fix, hardcoded column index map for Compare fields
15	BALIK_10.bas	VBA (Access)	IMPORT XLSX / BIBTEX / DIFF VIEWER buttons, Ctrl+B shortcut, report overflow fix
16	BALIK_11.bas	VBA (Access)	NOTES field in TAXONOMY tab, left sidebar navigation, CARDINAL filter-clear fix

Prerequisites

Before running any code, ensure the following software is installed and licensed on your Windows machine:

Microsoft Access 2016 or later (part of Microsoft Office)
Microsoft Excel 2016 or later
Microsoft Word 2016 or later — required for DOCX and PDF export functionality (Packages 3 and 5)
Google Colab account (free tier is sufficient)
Python 3 with pandas and xlsxwriter — only needed if running the Step 01 script locally instead of in Colab
Windows Script Host (WSH) — built into Windows, no installation needed

Place the following files in the same working folder before starting:

Your raw taxon data as an XLSX file (uploaded manually during Step 01)
vw_Complete_Taxa.xlsx — generated by Step 01, used by Step 02
All .wsf and .txt code files

> ⚠ **Enable VBA Macros Before Running Packages 1–5**
>
> Before running any VBA package, open Hyolith_Database.accdb and go to:
>
> `File → Options → Trust Center → Trust Center Settings → Macro Settings`
>
> Select "Enable all macros". Also add your database folder as a Trusted Location (Trust Center → Trusted Locations → Add new location). Without these settings Access will silently block all package build procedures.

01 Colab_create_vw_Complete_Taxa.txt

Google Colab · Python · pandas · xlsxwriter

Purpose

This Google Colab notebook transforms your raw flat XLSX data file into a structured multi-sheet Excel workbook called vw_Complete_Taxa.xlsx. This output file is the only input required by the next step (02).

What the Code Does

1. Installs xlsxwriter silently, then prompts you to upload your raw XLSX file via the Colab file picker.
2. Reads the uploaded file into a pandas DataFrame.
3. Assigns sequential integer IDs (column ID, starting at 1) and copies them to a TaxonID column used as a foreign key in all child sheets.
4. Detects any columns in your source data that are not listed in the known schema — these are collected as "unknown columns" and appended to the Taxa_Poznamky sheet so no data is lost.
5. Writes the output workbook with 7 sheets:

Field / Column	Description
`vw_Complete_Taxa`	Full flat view — all columns from the source, without TaxonID. Blue header row.
`Taxa`	Core taxonomy: ID, TAXON, Genus, Species, AUTHOR, Phylum, Family, Order, Class, Rank
`Taxa_Taxonomie`	Placement and hierarchy: TaxonID, TAXONOMIC PLACEMENT, Taxonomy_path, TYPE TAXON, INCLUDED TAXONS
`Taxa_Popis`	Morphological description: ETYMOLOGY, DESCRIPTION, SIZE, DIAGNOSIS, APERTURE, CROSS-SECTION, OPERCULUM, SCULPTURE, CLAVICLES, CARDINAL_PROCESSES
`Taxa_Material`	Specimens and literature: TYPE MATERIAL, MATERIAL EXAMINED, FIGURES, SYNONYMY
`Taxa_Vyskyt`	Occurrence and stratigraphy: OCCURRENCE, Geography, Locality, STRATIGRAPHY, System, Formation, Member, Stage, Serie, Zone, Horizon
`Taxa_Poznamky`	Remarks and references: REMARKS, Reference, plus any unrecognised source columns

6. Formats all header rows (blue for the flat view, green for the individual table sheets) and sets a default column width of 15 characters throughout.
7. Downloads the finished vw_Complete_Taxa.xlsx directly to your browser.

How to Run

```
# In Google Colab:
# 1. Open a new notebook
# 2. Paste the entire script into a code cell
# 3. Run the cell (Shift+Enter)
# 4. When prompted, upload your raw XLSX file
# 5. The output file downloads automatically
```

Key Functions

transformuj_a_stahni(df_vstup) — main pipeline: assigns IDs, splits into sheets, applies formatting, writes and downloads the file

Unknown-column detection — any source column not listed in the schema is silently preserved in Taxa_Poznamky instead of being discarded

02 code_create_database.wsf

Windows Script Host · VBScript · Access DAO · Excel COM

Purpose

This WSF script (Windows Script File) creates a brand-new Microsoft Access database file (Hyolith_Database.accdb), builds the full table structure, and imports all data from the vw_Complete_Taxa.xlsx workbook produced in Step 01. It must be run from the command line using cscript.exe.

What the Code Does

1. Detects whether it is running in the console (cscript.exe) or via wscript.exe, adjusting progress output accordingly.

2. Deletes any existing Hyolith_Database.accdb in the same folder (with error handling — aborts cleanly if Access has the file open).

3. Creates a new blank Access database via the Access.Application COM object and opens the Excel workbook via Excel.Application COM.

4. Executes CREATE TABLE DDL statements for all six tables with the exact column structure matching the Python script output:

Field / Column	Description
`Taxa`	LONG PRIMARY KEY (ID) + 9 LONGTEXT columns for core classification
`Taxa_Taxonomie`	TaxonID LONG (FK) + 5 LONGTEXT columns for hierarchical placement
`Taxa_Popis`	TaxonID LONG (FK) + 12 LONGTEXT columns for morphological description
`Taxa_Material`	TaxonID LONG (FK) + 4 LONGTEXT columns for specimens and literature
`Taxa_Vyskyt`	TaxonID LONG (FK) + 11 LONGTEXT columns for occurrence and stratigraphy
`Taxa_Poznamky`	TaxonID LONG (FK) + REMARKS, Reference, plus dynamic extra columns

5. Iterates over each of the 6 named sheets in the Excel file. For each sheet it reads the header row to build the field name array, then loops through all data rows, inserting each record with a parameterised INSERT INTO statement. Numeric fields (ID, TaxonID) are written without quotes; all other values use LONGTEXT quoting with apostrophe escaping.

6. Shows a real-time percentage progress counter in the console window (cscript mode only).

7. Writes every event and error to import_log.txt in the same folder, then closes both applications cleanly.

How to Run

```
# Open Command Prompt in the folder containing the WSF and XLSX files
cscript 02_code_create_database.wsf

# Monitor progress in the console window
# Check import_log.txt for the full record count and any errors
```

Key Functions

IsNumericField(tblName, fldName) — returns True only for ID and TaxonID columns; all other columns are treated as text

Log(msg) — writes timestamped messages to both console and import_log.txt

ShowProgress(current, total, sName) — prints a CR-based progress bar in cscript mode

03 code_access_table.txt

SQL · Microsoft Access · Query Designer

Purpose

This SQL statement creates the master view vw_Complete_Taxa inside Microsoft Access. It joins all six normalised tables into a single flat recordset, reproducing the original flat spreadsheet structure for forms, reports, and exports.

What the Code Does

Performs six cascading LEFT JOINs from the parent table Taxa outward to all five child tables using the TaxonID foreign key. Every column from every table is included in the SELECT list. Child TaxonID columns are aliased (Tx_TaxonID, Tp_TaxonID, etc.) to avoid name collisions.

The resulting query returns one row per taxon record with all fields accessible. Any taxon lacking a child record (e.g., no occurrence data) still appears in the result with NULL values in the missing columns.

How to Create This Query

Open Hyolith_Database.accdb in Microsoft Access

Go to Create > Query Design, close the table picker, switch to SQL View

Paste the entire SQL code and save the query with the name vw_Complete_Taxa

This name is hardcoded throughout all VBA form builder packages — do not rename it

Tables Joined

Field / Column	Description
`Taxa (t)`	Root table — ID is the primary key used in all joins
`Taxa_Taxonomie (tx)`	LEFT JOIN on t.ID = tx.TaxonID
`Taxa_Popis (tp)`	LEFT JOIN on t.ID = tp.TaxonID
`Taxa_Material (tm)`	LEFT JOIN on t.ID = tm.TaxonID
`Taxa_Vyskyt (tv)`	LEFT JOIN on t.ID = tv.TaxonID
`Taxa_Poznamky (tz)`	LEFT JOIN on t.ID = tz.TaxonID

04 create_column_compare.txt

Microsoft Access · Table Design View

Purpose

Adds a Boolean flag column named Compare to the Taxa table. This column is used by the comparison form (frmCompare4, built in Step 07) to mark individual records for side-by-side analysis and batch operations.

How to Add the Column Manually (Design View)

The Compare column is added directly in Access Table Design View — no SQL or code is needed. Follow these steps:

Field / Column	Description
`1. Open the table`	In the Navigation Pane (left sidebar), right-click the Taxa table and choose Design View.
`2. Scroll to the last row`	Click in the first empty Field Name cell below the last existing column.
`3. Enter the field name`	Type exactly: Compare (capital C, no spaces).
`4. Set the data type`	Click the Data Type dropdown in the same row and select Yes/No.
`5. Set the default value`	In the Field Properties panel at the bottom of the screen, find Default Value and type: No
`6. Save the table`	Press Ctrl+S. If Access asks to confirm, click Yes.
`7. Verify`	Switch to Datasheet View (Home → View → Datasheet View). A new checkbox column named Compare should appear at the right end of the table, with all checkboxes unchecked.

Alternative: SQL Method

If you prefer to use SQL, open a new query in SQL View (Create → Query Design → close the table picker → SQL View), paste the statement below, and click Run (the red ! button):

```
ALTER TABLE Taxa ADD COLUMN Compare YesNo;
```

Both methods produce identical results. The Design View method is recommended for users who are not comfortable with SQL.

Important Notes

The column name must be exactly Compare — it is referenced by name in the VBA packages (Packages 1, 2, and 4)

Do not use the name as a field filter in vw_Complete_Taxa manually — the query definition in Step 03 already excludes it from the GetFieldList() function

All existing records will have Compare = No (unchecked) after the column is added

05 code_create_table_history.txt

SQL · Microsoft Access

Purpose

Creates the Taxa_History table, which serves as an audit log for all field-level changes made through the Access GUI. This table powers the Undo functionality in the detail form and the History viewer in Package 4.

Table Structure

Field / Column	Description
`HistoryID`	AUTOINCREMENT PRIMARY KEY — unique identifier for each history record
`TaxonID`	LONG — ID of the taxon record that was changed
`TableName`	TEXT(255) — name of the source table (e.g., Taxa_Popis)
`FieldName`	TEXT(255) — name of the field that was modified
`OldValue`	MEMO — the previous value of the field before the change
`ChangeAction`	TEXT(50) — action type: UPDATE, DELETE, or INSERT
`ChangeDate`	DATETIME — timestamp of the change

How to Run

In Access, open a new query in SQL View

Paste the CREATE TABLE statement and execute it

The table is referenced by name in all VBA packages — do not rename it

Installing VBA Packages (Steps 06–11) — General Procedure

All six VBA files use the same installation method. Follow these steps for each package before running its specific build procedure:

Field / Column	Description
`1. Open VBA Editor`	Press Alt+F11 inside Access to open the Visual Basic Editor
`2. Insert a module`	Go to Insert → Module to create a new standard module (not a class module)
`3. Paste the code`	Copy the entire contents of the .txt file and paste into the module
`4. Run the procedure`	Place the cursor anywhere inside the public Sub (e.g. BuildForms_1_Zaklad) and press F5, or type its name in the Immediate Window (Ctrl+G) and press Enter
`5. Watch the output`	Progress messages appear in the Immediate Window. A message box confirms completion.
`6. Repeat in order`	Always install packages in the sequence 06 → 07 → 08 → 09 → 10 → 11. Some forms reference previously created ones (e.g. frmDetailTaxa expects frmTextPopup from Package 4 to exist).

06 code_main_form.txt

VBA · Microsoft Access · Package 1

Purpose

This is the first of five VBA form-builder packages. It constructs the main search interface of the database application. Running the entry procedure BuildForms_1_Zaklad creates two forms: frmSubResults (a scrollable datasheet subform) and frmDark (the main search and navigation form).

Visual Design

All forms use a unified dark theme with the following colour palette:

Field / Column	Description
`Background (C_BG)`	#202020 — near-black main background
`Panels (C_PANEL)`	#2B2B2B — slightly lighter panel areas
`Accent (C_ACCENT)`	#FF8C00 — orange highlight for titles and active elements
`Text (C_TEXT)`	#FFFFFF — white primary text
`Dim text (C_DIM)`	#C0C0C0 — grey secondary labels
`Borders (C_BORDER)`	#444444 — subtle dark grey borders

frmDark — Main Search Form

The main form provides multi-field live search with the following components:

Dual-column filter rows — each with two independent OR-joined input fields: TAXON, Genus, Family, Occurrence, Stratigraphy, Serie, Formation, Geography, Remarks, Operculum, Clavicles, Cardinal processes, Aperture, Cross-section, Sculpture

Dropdown combos for controlled vocabulary fields loaded from vw_Complete_Taxa: TAXON, Genus, Family, Serie, Formation, Operculum, Clavicles, Cardinal processes, Aperture, Cross-section, Sculpture

All combo boxes have LimitToList = False and AutoExpand = True — values can be typed freely or selected from the list

Free-text boxes for open-ended fields: Occurrence, Stratigraphy, Geography, Remarks

AND/OR logic toggle checkboxes between filter columns

CLEAR buttons beside every input to reset individual filters

Full-text SEARCH button that queries vw_Complete_Taxa and populates the subform

Record count label and active filter display updated after each search

Navigation buttons: Detail, Quick Preview, Compare, Export, New Taxon, Delete, Find/Replace, Bulk Edit, Bulk Rename, Validate, History, Backup DB, Batch DOCX, Import CSV

Keyboard Shortcuts (frmDark)

Field / Column	Description
`Esc`	Clear all filter fields
`Ctrl+N`	Open New Taxon form
`Ctrl+Z`	Undo last change
`Enter (on subform row)`	Open detail form for selected record
`Double-click (subform row)`	Open detail form for selected record
F4	Open Quick Preview (frmQuickPreview)
Ctrl+D	Open Detail form for the selected subform record
Ctrl+E	Export XLSX (same as EXPORT XLSX button)
Ctrl+B	Open BibTeX export form (frmBibTexExport)

Key Shared Functions (used by all packages)

SanitizeName(s) — strips all characters illegal in Access control names (spaces, brackets, slashes, special chars)

UpdateFieldInDB(tid, fldName, newVal) — routes a field update to the correct table based on field name, handles apostrophe escaping, executes UPDATE SQL

GetFieldList() — reads the vw_Complete_Taxa query definition and returns all field names except ID and Compare as a pipe-delimited string

How to Install

```
' In Access VBA Editor (Alt+F11):
' 1. Create a new standard module
' 2. Paste the entire contents of 06_code_main_form.txt
' 3. Open the Immediate Window (Ctrl+G)
' 4. Type: BuildForms_1_Zaklad
' 5. Press Enter — forms are built and saved automatically
```

07 code_compare_merge.txt

VBA · Microsoft Access · Package 2

Purpose

Package 2 builds frmCompare4, a modal popup form that displays up to four taxon records side by side in a scrollable panel. It is designed to support taxonomic decision-making: identifying synonyms, merging duplicate records, and selecting the preferred variant.

frmCompare4 — Side-by-Side Comparison Form

The form consists of:

A title bar with LOAD IDs, SAVE ALL, and CLOSE buttons

Four ComboBox controls (cboTaxon1–4) bound to vw_Complete_Taxa, allowing the user to select any taxon by name

A scrollable grid of labelled rows, one row per field, with one editable text column per selected taxon

Compact row height (320 twips) to maximise visible fields

Larger row height (700 twips) for long-text fields such as DESCRIPTION and DIAGNOSIS

Double-clicking any long-text field opens frmTextEditor — a popup text editor. See the Known Issues note below.

SAVE ALL writes all edited values back to the database across all four taxa simultaneously using UpdateFieldInDB

Supports merge workflow: called from the MERGE button on frmDark, pre-loading the two selected records

Modal, dark theme, ESC to close

> ⚠ **Known issue: popup editor in frmCompare4**
>
> ⚠ **frmTextEditor resolved by BALIK_8_FIXES**
>
> The OpenPopupEditor() function in frmCompare4 calls DoCmd.OpenForm "frmTextEditor". This form is created automatically by BALIK_8_FIXES (Step 13) with the correct name. After running InstallFixes8, the popup editor works in frmCompare4 without any manual renaming. The earlier workaround of renaming frmTextPopup to frmTextEditor in the Navigation Pane is no longer needed.

How to Install

```
' 1. Add code to existing module or create a new one
' 2. In Immediate Window type: BuildForms_2_Compare
' 3. Press Enter
```

Key Features

Horizontal scroll bar allows viewing beyond screen width when four wide columns are loaded

AutoExpand and LimitToList on combos ensure only valid taxa can be selected

Section height auto-expands as field rows are added during construction

08 code_export_import.txt

VBA · Microsoft Access · Package 3

Purpose

Package 3 creates five forms covering all data import and export operations: CSV import, XLSX import, DOCX export (single record), batch DOCX export, and BibTeX citation export.

Forms Created

Field / Column	Description
`frmImportCSV`	Imports taxon data from a delimited CSV file. Configurable separator (default comma), optional header row, optional duplicate TAXON skipping. Shows a 10-row preview before committing. Results are logged in an on-screen text box.
`frmImportXlsx`	Imports from an XLSX file. Sheet selector combo auto-populates from the chosen file. Supports column-to-field mapping (Excel columns → database fields), duplicate skipping, and the same preview/log workflow as CSV import. Uses ADO to read the file. Accessible via the IMPORT XLSX button in Row B of frmDark.
`frmExportDocx`	Exports the currently selected taxon as a formatted Word document (.docx) using Microsoft Word automation. Choose output folder and which sections to include. Requires Microsoft Word to be installed.
`frmBatchDocx`	Batch-exports multiple taxa (all currently visible, or only those flagged with Compare = Yes) to DOCX. Option for one file per taxon or a single combined document. Progress is shown in a log text box.
`frmBibTex`	Exports references in BibTeX, RIS, or plain text format. Scope options: current filter, selected taxon, or all records. Generates author/year citation keys automatically from the Reference field. Includes preview and save.

How to Install

```
' In Immediate Window: BuildForms_3_ImportExport
```

09 code_utilities.txt

VBA · Microsoft Access · Package 4

Purpose

Package 4 builds ten utility and analysis forms that support data quality, taxonomy management, and database maintenance.

Forms Created

Field / Column	Description
`frmFindReplace`	Global find-and-replace across any field in any table. Supports case-sensitive matching and whole-word-only mode. All replaced values are backed up to Taxa_History before overwriting. Preview before commit.
`frmValidator`	Data quality checker. Detects: missing mandatory fields, duplicate TAXON names, inconsistent Family/Order/Class assignments, records without geography or stratigraphy, records without any description. Results are listed and each entry can be clicked to open that record in the detail form.
`frmBulkRename`	Renames a taxonomic value across all records at once (e.g., correcting a misspelled formation name throughout the dataset). Shows affected records before applying. Backs up each change to Taxa_History.
`frmNewTaxon`	New-record wizard. Provides a minimal form for entering a new taxon with required fields and cascading dropdowns. Checks for duplicate TAXON name before saving. After creation, opens the detail form for the new record automatically.
`frmHistory`	Audit log viewer. Filterable by TaxonID and action type. Allows undoing selected history entries (restores OldValue to the database) or deleting history records. Also accessible from the HIST button in frmDetailTaxa.
`frmDiffViewer`	Field-level diff viewer. Compares the current database value of any field against a pasted reference text and highlights differences character by character. Useful for reviewing LLM extraction quality against the original publication text. Accessible via the DIFF VIEWER button in Row C of frmDark.

⚠ Note: Header comment vs. actual code

The comment block at the top of 09_code_utilities.txt lists ten forms including frmTranslate, frmStats, frmTreeView, and frmSavedFilters. These forms are NOT implemented in the current version of the code — BuildForms_4_Nastroje actually creates only the six forms listed above. This is a known discrepancy between the header documentation and the actual code.

Delete Taxon Logic (shared helper)

GetDeleteTaxonFunction() generates the VBA code string for the btnDeleteTaxon_Click event. When triggered:

Retrieves the selected taxon ID from the subform recordset

Asks for confirmation with the taxon name displayed

Calls BackupTaxonForDelete(tID) to write all field values to Taxa_History with ChangeAction = DELETE

Executes DELETE FROM on all six child tables in dependency order, then deletes from Taxa

Refreshes the subform and shows an Undo reminder

How to Install

```
' In Immediate Window: BuildForms_4_Nastroje
```

10 code_preview.txt

VBA · Microsoft Access · Package 5

Purpose

Package 5 builds two read-focused outputs: frmQuickPreview (a lightweight floating panel for rapid browsing) and rptTaxaDetail (a printable Access report).

frmQuickPreview — Fast Field Preview Panel

This non-modal floating form shows all key fields for the currently highlighted record in frmDark without opening the full detail editor. It is designed for rapid browsing through search results.

The form displays 22 labelled fields:

Header: taxon name in large orange font (18pt), Rank | Family | Order line beneath

Scrollable body with field labels and values for: Author, Placement, Tax. Path, Type Taxon, Included Taxons, Etymology, Size, Description, Diagnosis, Aperture, Cross-section, Operculum, Sculpture, Clavicles, Cardinal Processes, Occurrence, Geography, Stratigraphy, Locality, Synonymy, Type Material, Remarks

Keyboard shortcut (Escape) closes the form

Non-modal: the main search window remains active and focusable behind the preview panel

rptTaxaDetail — Printable Report

A formatted Access report that renders a single taxon record as a structured document suitable for printing or PDF export. Section order mirrors the publication format used in Hyolitha monographs: Taxonomy, Description, Occurrence, Material, Remarks.

How to Install

```
' In Immediate Window: BuildForms_5_PreviewReport
```

11 code_detail.txt

VBA · Microsoft Access · frmDetailTaxa v83

Purpose

This is the most complex component: the full record editor frmDetailTaxa. It provides a tabbed interface with inline editing for every field across all six tables, plus toolbar actions for saving, undoing, duplicating, and navigating records.

Form Architecture

The form is built programmatically at 31,000 twips wide with a 2,200-twip toolbar header and a 15,000-twip tabbed body. All layout constants are defined as Private Const values at the top of the module for easy adjustment.

Tabs and Field Layout

Field / Column	Description
`TAXONOMY`	TAXONOMIC PLACEMENT (read-only), Taxonomy_path (read-only), TAXON (editable), Rank, AUTHOR, Genus, Species, Family, Order, Class, Phylum, TYPE TAXON, SYNONYMY, INCLUDED TAXONS
`DESCRIPTION`	DESCRIPTION (large, editable), DIAGNOSIS (large), REMARKS (medium), OPERCULUM, APERTURE, CROSS-SECTION, SCULPTURE, CLAVICLES, CARDINAL_PROCESSES, ETYMOLOGY, SIZE
`OCCURRENCE`	OCCURRENCE, Geography, STRATIGRAPHY, Locality, Stage, Serie, System, Formation, Member, Zone, Horizon
`MATERIAL`	TYPE MATERIAL, MATERIAL EXAMINED, FIGURES, SYNONYMY, Taxonomy_path
`RELATIONSHIPS`	Shows other taxa sharing the same Genus or Family, with record counts. Clicking any entry opens that taxon directly in the detail form.
`NOTES`	REMARKS, Reference — plus any extra columns stored in Taxa_Poznamky

Field Layout Rules

Fields are defined using a compact string format: FieldName\isBig\isEdit\height

Field / Column	Description
`isBig = 1`	Full-width field spanning all columns
`isBig = 0`	Narrow field in a 3-column grid layout
`isBig = 2`	Half-width field spanning 2 columns
`isEdit = 1`	Editable text box
`isEdit = 0`	Read-only locked text box (grey background)
`height = N`	Normal height (600 twips)
`height = L`	Long text (3000 twips) — for DESCRIPTION, DIAGNOSIS
`height = R`	Medium text (1800 twips) — for REMARKS

Toolbar Buttons

Field / Column	Description
SAVE	Writes all modified fields to the database via SaveField(), which routes each value to the correct table and creates an undo point in Taxa_History
UNDO	Reads from Taxa_History to restore the last saved value of the most recently changed field
COPY / DUPLICATE	Creates a new taxon record copying all field values, then opens the new record for editing (TAXON name is left blank to force a unique name)
PREV / NEXT	Navigates to the previous or next record in the current search result set without closing the form
POPUP EDITOR	Opens frmTextPopup — a large resizable text editor for fields with long content. Double-clicking any long-text field also triggers this popup.
PDF	Exports the current taxon to a formatted PDF via Microsoft Word automation. Requires Word to be installed.
DOCX	Opens frmBatchDocx (Package 3) with the current taxon preselected for single-record export
HIST	Opens frmHistory filtered for the current TaxonID, showing the full change log for this record
DEDUP	Removes duplicate entries from pipe-separated fields such as SYNONYMY (e.g. if the same synonym appears twice)
CLOSE	Closes the form and returns focus to frmDark

Save and Backup Logic

SaveField(tid, fldName, newVal) first calls BackupFieldForUndo(tid, fldName) which writes the current database value to Taxa_History, then executes the UPDATE statement. This means every save automatically creates an undo point.

Table Routing

All field-to-table routing uses a SELECT CASE block on the uppercase field name. The routing is kept consistent with UpdateFieldInDB in Package 1, ensuring that forms built in different packages always write to the same table.

How to Install

```
' 1. In the VBA Editor, create a new standard module named modDetailTaxa
' 2. Paste the entire contents of 11_code_detail.txt
' 3. Run the detail form builder — in the Immediate Window type:
'       CreateDetailFormTaxa_Ultimate
'    Press Enter. A message box confirms: "Form v83 OK."
' 4. Run the popup editor builder separately — in the Immediate Window type:
'       CreateTextPopupForm
'    Press Enter. This creates frmTextPopup, used by the popup editor
'    button in frmDetailTaxa.
```

> ⚠ **Two procedures must be run separately**
>
> CreateDetailFormTaxa_Ultimate and CreateTextPopupForm are two independent Public Subs. The first does NOT call the second automatically. Both must be executed manually in the Immediate Window. If you skip CreateTextPopupForm, the popup editor button in frmDetailTaxa will fail at

runtime with an error that frmTextPopup cannot be found. Note: CreateTextPopupForm creates frmTextPopup for frmDetailTaxa. A separate form frmTextEditor (for frmCompare4) is created by BALIK_8_FIXES (Step 13) — you do not need to create it manually.

⚠ Known issue: frmCompare4 popup references frmTextEditor

The OpenPopupEditor() function embedded in frmCompare4 (Package 2) calls DoCmd.OpenForm "frmTextEditor" — but no procedure in the codebase creates a form with that name. frmTextPopup (created above) is a different form. If you click the popup editor inside frmCompare4, Access will report that frmTextEditor cannot be found. This issue is resolved by BALIK_8_FIXES (Step 13). After running InstallFixes8, frmTextEditor is created with the correct name and the popup works without any manual renaming.

Complete Build Workflow

Follow this exact sequence. Each step depends on the output of the previous one.

Field / Column	Description
`Step 01`	Run the Colab notebook. Upload your raw XLSX. Download vw_Complete_Taxa.xlsx.
`Step 02`	Place vw_Complete_Taxa.xlsx and 02_code_create_database.wsf in the same folder. Run: cscript 02_code_create_database.wsf. Verify import_log.txt shows no errors.
`Step 03`	Open Hyolith_Database.accdb in Access. Create a new query in SQL View. Paste code from 03_code_access_table.txt. Save as vw_Complete_Taxa.
`Step 04`	Create a new query. Paste code from 04_create_column_compare.txt. Execute once to add the Compare column.
`Step 05`	Create a new query. Paste code from 05_code_create_table_history.txt. Execute once to create Taxa_History.
`Step 06`	Open VBA Editor (Alt+F11). Create module. Paste 06_code_main_form.txt. In Immediate Window: BuildForms_1_Zaklad
`Step 07`	Create module. Paste 07_code_compare_merge.txt. In Immediate Window: BuildForms_2_Compare
`Step 08`	Create module. Paste 08_code_export_import.txt. In Immediate Window: BuildForms_3_ImportExport
`Step 09`	Create module. Paste 09_code_utilities.txt. In Immediate Window: BuildForms_4_Nastroje
`Step 10`	Create module. Paste 10_code_preview.txt. In Immediate Window: BuildForms_5_PreviewReport
`Step 11`	Create module named modDetailTaxa. Paste 11_code_detail.txt. In Immediate Window run: CreateDetailFormTaxa_Ultimate, then separately: CreateTextPopupForm. (frmTextEditor for Compare is created by BALIK_8 in Step 13 — no action needed here.)

Final Result — What You Get

After completing all 11 steps you will have a fully functional Microsoft Access database application with the following capabilities:

Field / Column	Description
`Search & Navigation`	Main form frmDark with multi-field filtering, live record count, keyboard shortcuts, and a scrollable datasheet subform
`Record Editing`	Tabbed detail form frmDetailTaxa covering all fields across 6 tables, with inline editing, popup text editor, and per-save undo points
`Comparison & Merge`	frmCompare4 for side-by-side editing of up to 4 taxa, with merge workflow integration
`Data Import`	CSV and XLSX import with column mapping, duplicate skipping, and preview
`Data Export`	Single and batch DOCX export via Word automation; PDF export; BibTeX and RIS citation export
`Data Quality`	Validator (missing fields, duplicates, inconsistencies), Find & Replace with history backup, Bulk Rename
`Change History & Undo`	Full field-level audit log in Taxa_History; undo from both the detail form and the standalone history viewer
`Reference Tools`	Relationships tab showing taxonomically related records; Diff Viewer for comparing against source text
`Quick Preview`	Non-modal frmQuickPreview for rapid browsing; printable report rptTaxaDetail for publication-style output
`Dark UI Theme`	Consistent dark interface (#202020 background, #FF8C00 orange accents) across all forms

Database Schema Reference

The following diagram shows the relational structure. All five child tables link to Taxa via a TaxonID = ID foreign key.

```
Taxa (ID PK)
    ├── Taxa_Taxonomie  (TaxonID FK)
    ├── Taxa_Popis      (TaxonID FK)
    ├── Taxa_Material   (TaxonID FK)
    ├── Taxa_Vyskyt     (TaxonID FK)
    ├── Taxa_Poznamky   (TaxonID FK)
    └── Taxa_History    (TaxonID FK — audit log)
```

The query vw_Complete_Taxa flattens this structure back into a single row per taxon for use in all forms and exports.

Troubleshooting

Field / Column	Description
`WSF script aborts with 'Close Access'`	The .accdb file is open in Access. Close Access completely before running the import script.
`Import log shows 0 records`	Check that the sheet names in vw_Complete_Taxa.xlsx exactly match: Taxa, Taxa_Taxonomie, Taxa_Popis, Taxa_Material, Taxa_Vyskyt, Taxa_Poznamky.
`WSF script expects exact filename`	The script looks for vw_Complete_Taxa.xlsx in the same folder. Do not rename the Excel output from Step 01.
`Form build fails with 'Object not found'`	The query vw_Complete_Taxa has not been created yet, or was saved with a different name. Complete Step 03 first.
`BuildForms_ procedure not found`	You may have pasted the code into a class module instead of a standard module. Delete it and create a new standard module (Insert > Module).
`VBA macros blocked silently`	Macros are disabled by Trust Center settings. See the Prerequisites section — enable macros and add your folder as a Trusted Location.
`PDF or DOCX export fails`	Microsoft Word must be installed. All export forms use late binding (CreateObject("Word.Application")), so no VBA References need to be set manually — but Word itself must be present.
`Popup editor in frmDetailTaxa fails`	CreateTextPopupForm must be run separately after CreateDetailFormTaxa_Ultimate. Both are independent Public Subs in Step 11 — the first does not call the second automatically.
`Popup editor in frmCompare4 fails with 'frmTextEditor not found'`	Resolved by BALIK_8_FIXES (Step 13). Run InstallFixes8 — frmTextEditor is created automatically with the correct name. No manual renaming is needed.
`Controls invisible or overlapping`	Screen resolution or Access DPI scaling may differ. Adjust the layout constants (FORM_W, GRID_TXT_W, etc.) at the top of the relevant module.
`VBA compile error on CROSS-SECTION`	Access does not allow hyphens in field names in some contexts. Use square brackets: [CROSS-SECTION] in any SQL you write manually.
`Any form fails to create`	Open the Immediate Window (Ctrl+G) immediately after the error — Access usually prints a descriptive message there. Common causes: missing table, misspelled field name, or a previous partially-built form that needs to be deleted first (DoCmd.DeleteObject acForm, "frmName").

Patch Packages — Installation Guide

After completing the standard build (Steps 01–11), three additional VBA patch modules must be installed to fix known runtime errors and add new functionality. These patches are independent of the original build sequence and can be applied to any existing Hyolith_Database.accdb that has already had Steps 06–11 completed.

What the Patches Fix and Add

Package	Description
BALIK_6_7.bas	Main combined patch: fixes three runtime errors (missing forms, SQL syntax), adds filter presets, scope-aware export and bulk edit, and improved Compare Save All feedback
BALIK_8.bas	Secondary patch: fixes a VBA compile error in frmMergePreview, creates the missing frmTextEditor form, and fixes silent data loss when loading CROSS-SECTION and CARDINAL_PROCESSES fields in frmCompare4

Prerequisites

Both patches require that Steps 06–11 of the standard build have already been completed successfully. The following forms and tables must already exist:

frmDark, frmCompare4, frmValidator, frmDetailTaxa, frmTextPopup, taxa table, Taxa_History table, vw_Complete_Taxa query.

Installation Order

⚠ Always install BALIK_6_7 before BALIK_8. The second patch patches frmMergePreview which is created by the first.

Order	Action
Step 1	Import BALIK_6_7.bas into the VBA Editor and run InstallAll
Step 2	Import BALIK_8.bas into the VBA Editor and run InstallFixes8
Step 3	Save the database (Ctrl+S in Access)

Installation Procedure (applies to both files)

1. Open Microsoft Access and open Hyolith_Database.accdb.
2. Press Alt+F11 to open the Visual Basic Editor.
3. Go to File → Import File and select the .bas file.

   The module is imported automatically with the correct name (BALIK_6_7 or BALIK_8).
4. Press Ctrl+G to open the Immediate Window.
5. Type the entry procedure name and press Enter:

```
InstallAll            ' for BALIK_6_7.bas
InstallFixes8         ' for BALIK_8.bas
```

6. Watch the Immediate Window — each step prints OK when complete.
7. A message box confirms successful installation.
8. After both patches are installed, save the database in Access (Ctrl+S).

BALIK_6_7— Detailed Description

New forms created

Form	Purpose
frmBulkFieldPicker	Field selection dialog for Bulk Edit. Lists all editable fields from vw_Complete_Taxa. User selects a field and enters a new value. Communicates with frmDark via TempVars.
frmMergePreview	Confirmation dialog shown before merging records. Displays which record will be kept (KEEP) and which will be deleted (DELETE). Reads merge parameters from frmDark global variables.
frmSavedFilters	Filter preset manager. Lists all saved filter presets with name and date. Allows saving the current filter state, loading a saved preset, and deleting presets. Presets are stored in the FilterPresets table.

New table creates

Table	Description
FilterPresets	Stores named filter presets. Columns: ID (AutoNumber PK), FilterName (Text 100), FilterData (Memo — serialised control values), SavedAt (Date). Created automatically on first run; skipped if already exists.

Bugs fixed

Bug	Fix
frmBulkFieldPicker not found	The form was never created in the original build. BALIK_6_7 creates it.
frmMergePreview not found	Same as above — the form was referenced but never built.
frmValidator: Syntax error in SQL	COUNT(DISTINCT field) is not valid in Access SQL. Replaced with a DAO cursor loop using DCount to find genera with inconsistent Family assignments. Also fixes a secondary error where JOIN on Memo fields (Genus, Family) fails in Access — DCount with Left() is used instead.

New functionality in frmDark

Feature	Description
SAVE FILTER button	Serialises all current filter control values (txt*, cbo*, chk* prefixes) into a pipe-delimited string and opens frmSavedFilters to name and save the preset.
LOAD FILTER button	Opens frmSavedFilters to select a saved preset. After selection, DeserializeFilters() restores all control values and ApplyFilters() executes the search automatically.
Export XLSX — scope aware	Now checks for checked (Compare = True) records first. If any are checked, exports only those. If none are checked, exports the current filtered set. If no filter is active, exports all records. A confirmation dialog shows the scope before proceeding.
Bulk Edit — scope aware	Same priority logic as Export: checked records first, then filtered, then all. A confirmation dialog shows field name, new value, and scope before any changes are made. Each affected record is backed up to Taxa_History before the UPDATE is executed.

Improved Compare Save All:

The btnSaveAll_Click handler in frmCompare4 now shows a detailed confirmation message after saving, listing the names of all updated taxa, the total number of fields written, and a reminder that changes are logged in Taxa_History and can be reverted with UNDO [Ctrl+Z] in the main search form. Previously the button showed only a generic "Saved." message with no information about what was actually written.

BALIK_8 — Detailed Description

Bugs fixed

Bug	Fix
frmMergePreview: End If without block If (compile error)	The inline single-line If...Else...End If syntax used in btnConfirm_Click and btnCancel_Click is not valid VBA when both branches contain statements. All If blocks are rewritten as proper multi-line If / Else / End If blocks. frmMergePreview is fully rebuilt.
frmTextEditor not found (runtime error 2102 in frmCompare4)	frmCompare4 calls DoCmd.OpenForm "frmTextEditor" when the user clicks any long-text field, but no procedure in the codebase created a form with that name. BALIK_8 creates frmTextEditor as a proper popup text editor with a large text area, character counter, and SAVE / CANCEL buttons communicating via TempVars(PopupText) and TempVars(PopupSaved).
CROSS-SECTION and CARDINAL_PROCESSES show empty in frmCompare4	The btnLoad_Click handler accessed recordset fields by name: rs.Fields("CROSS-SECTION"). DAO in Access treats the hyphen as a minus operator in some parsing contexts, causing the lookup to fail silently under On Error Resume Next. Fix: the field lookup is rewritten to iterate rs.Fields by numeric index and match on .Name — this approach is immune to special characters in field names.

New form created

Form	Description
frmTextEditor	Popup text editor for frmCompare4. Shows which field is being edited in the title label. Large editable text area (18000 x 7000 twips) with vertical scrollbar and Enter key enabled. Live character count. SAVE writes to TempVars(PopupText) and sets PopupSaved = "1". CANCEL sets PopupSaved = "0". ESC key triggers Cancel. This form replaces the workaround of manually renaming frmTextPopup described in the original build guide — both the detail form (frmDetailTaxa via frmTextPopup) and the comparison form (frmCompare4 via frmTextEditor) now have their own correctly named popup editors.

Updated Troubleshooting Entries

The following entries supersede or supplement the Troubleshooting table in the main build guide:

Problem	Solution
Bulk Edit error: frmBulkFieldPicker not found	Run InstallAll from BALIK_6_7.bas. The form was missing from the original build.
Merge error: frmMergePreview not found	Run InstallAll from BALIK_6_7.bas.
Validator: Syntax error — COUNT(DISTINCT ...)	Run InstallAll from BALIK_6_7.bas. frmValidator is rebuilt with Access-compatible SQL.
Validator: Memo field JOIN error	Run InstallAll from BALIK_6_7.bas. The Inconsistent Family check now uses a DAO cursor with DCount instead of a JOIN.
Merge Preview: compile error End If without block If	Run InstallFixes8 from BALIK_8.bas. frmMergePreview is rebuilt with correct VBA syntax.
Compare: clicking a field gives error frmTextEditor not found	Run InstallFixes8 from BALIK_8.bas. This creates frmTextEditor. The manual rename workaround (frmTextPopup → frmTextEditor) described in the original guide is no longer needed.
Compare: CROSS-SECTION and CARDINAL_PROCESSES fields always empty after Load	Run InstallFixes8 from BALIK_8.bas. The btnLoad_Click handler is patched to use index-based field lookup.
SAVE FILTER / LOAD FILTER buttons not visible	Run InstallAll from BALIK_6_7.bas. Buttons are added to Row C of frmDark (next to VALIDATE). If they appear at Y=0, switch frmDark to Design View and drag them next to the VALIDATE button.
Filter presets not saving	Verify that the FilterPresets table exists (check in the Access Navigation Pane under Tables). If missing, run InstallAll from BALIK_6_7.bas — it creates the table automatically.
Compile error: variable name mod	This affects older versions of the patch files where the variable name mod was used (mod is a reserved word in VBA). Use BALIK_6_7.bas from the current session — it uses mdl throughout.

Updated Complete Build Workflow

The table below shows the full sequence including the patch packages. Steps 01–11 are unchanged; Steps 12 and 13 are new.

Step	Action
01–11	Follow the original build sequence as documented in the main sections of this guide.
12 — BALIK_6_7	In VBA Editor: File → Import File → BALIK_6_7.bas. Immediate Window: InstallAll. Installs: frmBulkFieldPicker, frmMergePreview, frmSavedFilters, fixed frmValidator, FilterPresets table, patched frmDark (Export/BulkEdit scope + SAVE/LOAD FILTER buttons), improved frmCompare4 Save All.
13 — BALIK_8	In VBA Editor: File → Import File → BALIK_8.bas. Immediate Window: InstallFixes8. Installs: fixed frmMergePreview (compile error), frmTextEditor (popup editor for Compare), patched frmCompare4 btnLoad_Click (CROSS-SECTION / CARDINAL_PROCESSES fix).
Final	Save the database (Ctrl+S). Open frmDark and verify: filter works, SAVE FILTER and LOAD FILTER buttons are visible in Row C, Bulk Edit shows scope confirmation, Compare loads all fields including CROSS-SECTION and CARDINAL_PROCESSES, clicking a long-text field in Compare opens frmTextEditor.

BALIK_9 — Additional Patch

BALIK_9.bas addresses two bugs not covered by the previous patches. Install it after BALIK_8_FIXES using the same procedure (File → Import File in VBA Editor, then InstallFixes9 in the Immediate Window).

What BALIK_9 Fixes

Bug	Fix
frmBulkRename "Preview error: property setting is too long"	btnPreview_Click was building a semicolon-delimited Value List string for lstAffected.RowSource. Access hard-limits this string to ~32,700 characters. With hundreds of matching records (e.g. Class = Hyolithomorpha) the limit is exceeded. Fix: preview now saves a temporary QueryDef named qryBulkPreview and binds lstAffected to it via RowSourceType = Table/Query. No string length limit applies.
frmCompare4: CROSS-SECTION and CARDINAL_PROCESSES always empty after Load	All previous patches tried to resolve these fields by iterating rs.Fields and matching .Name. Access DAO does not reliably return the bare field name for hyphenated columns (CROSS-SECTION) sourced from a child table via JOIN in a saved QueryDef — the lookup fails silently under On Error Resume Next. Fix: btnLoad_Click now uses a hardcoded column index array derived from the confirmed SELECT column order in vw_Complete_Taxa. CROSS-SECTION is always at index 23, CARDINAL_PROCESSES at index 27. rs.Fields(idx(fi)).Value bypasses all name resolution.

Updated Complete Build Workflow (including BALIK_9)

Step	Action
01–11	Standard build sequence — unchanged.
12 — BALIK_6_7	Import BALIK_6_7.bas → InstallAll. Creates frmBulkFieldPicker, frmMergePreview, frmSavedFilters, fixes frmValidator, creates FilterPresets table, patches frmDark (scope-aware Export/BulkEdit, SAVE/LOAD FILTER buttons), improves Compare Save All feedback.
13 — BALIK_8	Import BALIK_8.bas → InstallFixes8. Fixes frmMergePreview compile error, creates frmTextEditor, patches frmCompare4 btnLoad_Click.
14 — BALIK_9	Import BALIK_9.bas → InstallFixes9. Fixes frmBulkRename preview crash, fixes CROSS-SECTION and CARDINAL_PROCESSES loading in frmCompare4 using hardcoded column index map.
15 — BALIK_10	Import BALIK_10.bas → InstallBALIK10. Adds three buttons (IMPORT XLSX, BIBTEX, DIFF VIEWER), removes dead btnCompare handler, adds Ctrl+B shortcut, fixes frmImportXlsx column mapping display (#NÁZEV? issue), fixes rptTaxaDetail section height overflow.
16 — BALIK_11	Import BALIK_11.bas → InstallBALIK11. Adds NOTES field to TAXONOMY tab (unbound, height 2400 twips, stored in Taxa_Poznamky). Adds [S] save button beside NOTES. Fixes CARDINAL_PROCESSES display after filter clear. Hides native tab strip and adds left sidebar navigation. Then run: CreateDetailFormTaxa_Ultimate.
Final	Save the database (Ctrl+S). Verify: Bulk Rename preview works without error on any field, Compare loads all 40 fields including CROSS-SECTION and CARDINAL_PROCESSES correctly, NOTES field visible in TAXONOMY tab with sidebar navigation working.

Complete Feature Reference

This section provides a complete reference for all features visible in the running application, supplementing the build instructions in the preceding sections.

frmDark — Button Reference

The main search form frmDark has three rows of action buttons below the filter panel. All buttons are accessible by mouse; some have keyboard shortcuts.

Row A — Search and Edit

Button	Function
FILTER	Applies all active filter fields and refreshes the subform recordset. The status dot turns blue when a filter is active; the active filter expression is shown in the status bar.
CLEAR [Esc]	Clears all filter fields, resets all Compare checkboxes in the database (UPDATE Taxa SET Compare = False), and re-runs the search to show all records.
NEW [Ctrl+N]	Opens frmNewTaxon — a minimal entry form for creating a new taxon record. After saving, frmDetailTaxa opens automatically for the new record.
UNDO [Ctrl+Z]	Reads the most recent entry from Taxa_History for the currently selected record and restores the previous value. Affects only the last single-field change.
PREVIEW [F4]	Opens frmQuickPreview — a non-modal floating panel showing all key fields for the selected record without opening the full detail editor.
FIND/REPLACE	Opens frmFindReplace — global find and replace across any field in any table. Supports case-sensitive and whole-word modes. Backs up all changed values to Taxa_History before applying.
EXPORT XLSX	Exports to Export.xlsx in the database folder. Scope priority: checked taxa (Compare = True) first; if none checked, exports current filtered set; if no filter, exports all records. Confirmation dialog shows the scope before proceeding.
BULK EDIT	Opens frmBulkFieldPicker to select a field and new value. Applies the value to all checked taxa, or to filtered taxa if none are checked. Shows scope and field/value in a confirmation dialog before executing. Backs up all affected records to Taxa_History.
BULK RENAME	Opens frmBulkRename — field-wide value rename (e.g. rename one Family value across all records). Select a field, choose the current value from a dropdown, enter the new value, preview affected records, then apply.

Row B — Analysis and Tools

Button	Function
COMPARE	Collects IDs of all checked (Compare = True) records and opens frmCompare4 with those taxa pre-loaded for side-by-side comparison.
MERGE	Collects checked record IDs and opens frmMergePreview. The first checked record is kept (KEEP); all others are deleted and their child table records are moved to the kept record. Backed up before execution.
UNDO MERGE	Reverses the most recent merge operation by reading the merge backup from Taxa_History and restoring deleted records.
DELETE	Deletes the currently selected record from all six tables after confirmation. The full record is backed up to Taxa_History with ChangeAction = DELETE before removal.
IMPORT CSV	Opens frmImportCSV. Configurable delimiter, optional header row, duplicate TAXON skipping, 10-row preview before committing.
BATCH DOCX	Opens frmBatchDocx. Exports checked taxa (or all filtered taxa if none checked) as formatted Word documents — one file per taxon or a single combined file.
IMPORT XLSX	Opens frmImportXlsx. Sheet selector auto-populates from the chosen file. Supports column-to-field mapping, duplicate skipping, and 10-row preview. Uses ADO to read the file without needing Excel installed.
BIBTEX	Opens frmBibTexExport. Exports references in BibTeX, RIS, or plain text format. Scope: current filter, selected taxon, or all records. Citation keys are generated automatically from the Reference field.

Row C — Views and Maintenance

Button	Function
HISTORY	Opens frmHistory — filterable audit log of all field-level changes. Each entry shows TaxonID, field name, old value, action type, and timestamp. Individual entries can be undone (restores OldValue) or deleted.
VALIDATE	Opens frmValidator — data quality checker. Detects missing required fields, duplicate TAXON names, inconsistent Family/Order/Class assignments (same genus, different values), records without geography or stratigraphy, and records without description. Each result can be clicked to open that record in the detail form.
SAVE FILTER	Serialises all active filter control values and opens frmSavedFilters to name and save the preset. Presets are stored in the FilterPresets table.
LOAD FILTER	Opens frmSavedFilters to select a saved preset. After selection, all filter controls are restored and the search is re-run automatically.
BACKUP DB	Creates a timestamped copy of Hyolith_Database.accdb in the same folder (e.g. Hyolith_Database_backup_20260313_223300.accdb). No Access automation required — uses FileCopy.
DIFF VIEWER	Opens frmDiffViewer. Compares the current database value of any field against pasted reference text and highlights differences character by character. Useful for reviewing LLM extraction accuracy against the original publication.

Subform and DETAIL Button

The right panel of frmDark contains frmSubResults — a read-only datasheet showing all columns from vw_Complete_Taxa for the current filter. The Compare column shows checkboxes that can be ticked directly in the datasheet. All other columns are read-only in this view.

The DETAIL button (or double-clicking any row, or pressing Enter) opens frmDetailTaxa for the selected record. frmDetailTaxa inherits the current filter as its recordset so PREV/NEXT navigation stays within the filtered set.

frmDetailTaxa — Complete Tab and Button Reference

frmDetailTaxa is a 31,000-twip-wide tabbed form with a 2,200-twip toolbar header. The record is identified by the title (TAXON AUTHOR, Year) and the ID / Rank line below it.

Tabs

Tab	Fields shown
TAXONOMY	TAXONOMIC PLACEMENT (read-only), Taxonomy_path (read-only), TAXON, Rank, AUTHOR, Genus, Species, Family, Order, Class, Phylum, TYPE TAXON, SYNONYMY, INCLUDED TAXONS
DESCRIPTION	DESCRIPTION, DIAGNOSIS, REMARKS, OPERCULUM, APERTURE, CROSS-SECTION, SCULPTURE, CLAVICLES, CARDINAL_PROCESSES, ETYMOLOGY, SIZE
OCCURRENCE	OCCURRENCE, Geography, STRATIGRAPHY, Locality, Stage, Serie, System, Formation, Member, Zone, Horizon
MATERIAL	TYPE MATERIAL, MATERIAL EXAMINED, FIGURES, SYNONYMY, Taxonomy_path
RELATIONSHIPS	Read-only tab. Shows other taxa that share the same Genus or Family as the current record, with record counts. Clicking any entry opens that taxon directly in frmDetailTaxa.
NOTES	REMARKS and Reference fields, plus any extra columns stored in Taxa_Poznamky for unrecognised source data. These fields are editable. Extra columns appear here automatically if the source data contained columns not in the standard schema.

Toolbar buttons

Button	Function
PDF	Exports the current record to a formatted PDF using Microsoft Word automation. Requires Word. Generates a structured document matching Hyolitha monograph format.
DOCX	Opens frmBatchDocx with the current taxon preselected for single-record Word export.
< PREV / NEXT >	Navigates to the previous or next record within the current filtered recordset, without closing the form. The record count (e.g. 31) in the centre field shows the current ID.
UNDO	Restores the last saved value for the most recently changed field by reading from Taxa_History.
COPY	Duplicates the current record: creates a new taxon in all six tables copying all field values, leaves TAXON blank (must be filled in), and opens the new record for editing.
HIST	Opens frmHistory filtered to the current TaxonID — shows the full field-level change log for this record.
DEDUP	Removes duplicate entries from pipe-separated fields such as SYNONYMY. For example, if the same synonym appears twice in a pipe-delimited list, DEDUP collapses it to one occurrence. Operates on all relevant fields of the current record.
X CLOSE	Closes frmDetailTaxa and returns focus to frmDark.

Save behaviour

Every field edit in frmDetailTaxa requires an explicit click on SAVE to persist. The SAVE button calls SaveField() for each modified field, which first calls BackupFieldForUndo() — writing the current database value to Taxa_History — then executes the UPDATE statement. This means every save automatically creates an undo point. There is no auto-save.

Long-text fields (DESCRIPTION, DIAGNOSIS, REMARKS, etc.) have a green + button that opens frmTextPopup — a large resizable popup editor — for comfortable editing. Double-clicking the field also opens the popup.

frmCompare4 — Comparison Form Reference

frmCompare4 opens as a modal popup. It can be launched from the COMPARE button in frmDark (pre-loads checked taxa) or from the MERGE button (pre-loads the two records to be merged).

Control / Button	Function
cboTaxon1–4	ComboBox controls bound to vw_Complete_Taxa. Select any taxon by typing or choosing from the dropdown. Changing a combo updates the label below it with the taxon name.
LOAD IDs	Reads all four combo values and fills the comparison grid by opening a recordset on vw_Complete_Taxa for each taxon. All 40 fields are populated using hardcoded column indices (immune to hyphen/space in field names).
SAVE ALL	Writes all edited values back to the database for all loaded taxa using UpdateFieldInDB. Shows a detailed confirmation with taxon names, field count, and a reminder that changes are logged in Taxa_History and can be reverted with UNDO.
CLOSE / ESC	Closes the form without saving.
Field cells	All 40 fields are displayed as editable text boxes. Long-text fields (DESCRIPTION, DIAGNOSIS, etc.) open frmTextEditor on click for comfortable editing in a popup window.

Automatic Database Backup

The BACKUP DB button in Row C of frmDark creates a timestamped copy of the database file in the same folder as the original. The filename format is:

Hyolith_Database_backup_YYYYMMDD_HHMMSS.accdb

The backup is a complete file-level copy (FileCopy) — it does not require Access DAO or COM automation and does not interfere with the open database. The copy is immediately usable as a standalone Access database. In addition to the manual button, frmDark automatically creates a backup on open via AutoBackup_OnOpen(). This runs once per calendar day: it checks whether a file matching Hyolitha_YYYYMMDD*.accdb already exists in the Backup\ subfolder and skips silently if it does. Backup files are stored in a Backup\ subfolder next to the main database, named Hyolitha_YYYYMMDD_HHNNSS.accdb. The folder is created automatically if it does not exist.

Filter Presets (SAVE FILTER / LOAD FILTER)

Filter presets store the complete state of all filter controls in frmDark as a pipe-delimited string in the FilterPresets table. The string encodes control names and values for all txt*, cbo*, and chk* prefixed controls that have non-empty / non-zero values.

Action	Description
SAVE FILTER	Serialises current filter state via SerializeFilters(), opens frmSavedFilters. Enter a name and click SAVE. If a preset with the same name already exists, you are asked whether to overwrite.
LOAD FILTER	Opens frmSavedFilters showing all saved presets with name and date. Select a preset and click LOAD SELECTED. DeserializeFilters() restores all control values; ApplyFilters() re-runs the search. The active filter label shows [Preset: name].
DELETE	In frmSavedFilters, select a preset and click DELETE. Asks for confirmation.
FilterPresets table	Underlying storage: ID (AutoNumber), FilterName (Text 100), FilterData (Memo), SavedAt (Date). The table is created automatically by BALIK_6_7.

Scope Logic for Bulk Operations

Export XLSX, Bulk Edit, and Batch DOCX all follow the same scope priority:

Priority	Condition
1 — Checked	If any records have Compare = True (checkbox ticked in the subform), the operation applies only to those records.
2 — Filtered	If no records are checked but a filter is active, the operation applies to the current filtered record set (the RecordSource of frmSubResults).
3 — All	If no records are checked and no filter is active, the operation applies to all records in vw_Complete_Taxa. For Bulk Edit this triggers an additional warning in the confirmation dialog.

For Bulk Edit and Export, a confirmation dialog always shows the resolved scope and the action (field + new value for Bulk Edit; destination file for Export) before any changes are made.

Data Validator — Check Types

Check	What it detects
Missing required fields	Records where TAXON, AUTHOR, or Rank is empty or NULL.
Duplicate TAXON names	Records sharing an identical TAXON string. Uses DCount to avoid Memo field limitations.
Inconsistent Family/Order/Class	Records where the same Genus appears with different Family values across different records. Uses a DAO cursor with DCount and Left(field, 100) comparison — immune to Memo field JOIN restrictions. Note: this check can be slow on large datasets (>1,000 records) because it runs a DCount query for each record individually.
Records without Geography and Stratigraphy	Taxa where both Geography and STRATIGRAPHY in Taxa_Vyskyt are empty.
Records without Description	Taxa where DESCRIPTION in Taxa_Popis is empty.

Each result in the list can be clicked to highlight the record, then OPEN DETAIL opens it in frmDetailTaxa for editing.

Relationships Tab in frmDetailTaxa

The RELATIONSHIPS tab shows two groups of related taxa for the current record:

Same Genus — all other records sharing the same Genus value, with total count.

Same Family — all other records sharing the same Family value, with total count.

Clicking any entry in the list opens that taxon directly in frmDetailTaxa, replacing the current record. This allows rapid navigation between related taxa without returning to the main search form.

DEDUP — Pipe-Separated Field Deduplication

Many fields in the database store multiple values as pipe-separated strings (e.g. SYNONYMY, INCLUDED TAXONS, OCCURRENCE). The DEDUP button in frmDetailTaxa scans all such fields for the current record and removes duplicate entries while preserving order.

Example: "Hyolithus mutatus | Hyolithus varians | Hyolithus mutatus" becomes "Hyolithus mutatus | Hyolithus varians".

DEDUP operates on the current record only. Each modified field is backed up to Taxa_History before the deduplication value is written. The operation is irreversible via UNDO only if the backup was created — verify in HIST after running DEDUP if you need to confirm the backup exists.

Problem	Solution
Bulk Rename: "Preview error: The property setting is too long"	Run InstallFixes9 from BALIK_9.bas. The form is rebuilt to use a QueryDef for the preview list instead of a Value List string.
Compare: CROSS-SECTION / CARDINAL_PROCESSES still empty after BALIK_8	Run InstallFixes9 from BALIK_9.bas. The fix uses hardcoded column indices (23 and 27) instead of name-based field lookup, which is unreliable for hyphenated field names in DAO.
Validator: Inconsistent Family check is very slow	Expected behaviour on datasets > 500 records. The check runs one DCount per record to avoid the Memo JOIN limitation. On 1,500 records this takes approximately 30–60 seconds. Use the check selectively.
Backup DB creates no file	Verify the database file is named exactly Hyolith_Database.accdb and is located in a folder where the current user has write permission. The backup uses the Access CurrentProject.FullName path.
RELATIONSHIPS tab shows no related taxa	The current taxon may have no other records sharing the same Genus or Family. Also check that the Genus and Family fields are not empty for the current record.
DEDUP button removes entries that should be kept	DEDUP performs case-sensitive string matching. Two entries that differ only in capitalisation or spacing are treated as distinct. Review the field value in the popup editor before running DEDUP on important records.

BALIK_10 — Additional UI Completeness Patch

BALIK_10.bas completes the frmDark button set by adding four forms that existed since Packages 3–4 but had no UI access point, removes a dead code handler, adds two keyboard shortcuts, and redesigns the frmDetailTaxa navigation sidebar. Install after BALIK_9 using the standard procedure (File → Import File, then InstallBALIK10 in the Immediate Window).

What BALIK_10 Changes

Change	Description
3 new buttons in frmDark	Row B: IMPORT XLSX (opens frmImportXlsx) and BIBTEX (opens frmBibTexExport). Row C: DIFF VIEWER (opens frmDiffViewer). All three forms existed since Packages 3–4 but had no UI access point before this patch. The dead btnCompare_Click handler (superseded by btnCompare4) is also removed.
Dead handler removed	btnCompare_Click (the original two-taxon compare handler, superseded by btnCompare4_Click in Package 2) is deleted from the frmDark module. It had no button and was never callable from the UI.
New keyboard shortcuts	Ctrl+B opens BibTeX export (frmBibTexExport). Ctrl+D (Detail) and Ctrl+E (Export XLSX) were already in the code but undocumented; they are preserved and now listed in the keyboard shortcuts table.
frmDetailTaxa navigation redesigned	The native Access tab strip is hidden and replaced by a vertical sidebar of six pill-style labels: TAXONOMY, DESCRIPTION, OCCURRENCE, MATERIAL, NOTES, RELATIONSHIPS. The active pill is highlighted in orange (#FF8C00). Clicking a pill calls SetFocus on the first field of that section, which causes Access to scroll the form body to that position. This approach avoids the unreliable show/hide of tab pages and gives each section its full natural height without clipping.

Navigation Sidebar — Design Notes

The Access native tab control has two limitations for this database: the tab strip is too narrow to display all six tab labels without truncation, and tab page heights cannot be adjusted to match the content of each section. The BALIK_10 sidebar uses a different approach:

1. The native tab control (tabMain) is set to Visible = False. The content controls remain on their tab pages but the tab strip disappears.

2. Six Label controls named lblNav_TAXONOMY through lblNav_RELATIONSHIPS are placed in a 1,700 × 560 twip column on the left side of the form, below the toolbar.

3. Each label's OnClick event calls NavHighlight() to update colors, then SetFocus on the first editable control in that section. Access scrolls the form body automatically to bring that control into view.

4. The active pill uses BackColor = #FF8C00 (orange) with black text. Inactive pills use BackColor = #444444 with white text. Colors update on every click via NavHighlight(), which iterates all six pills.

Keyboard Shortcuts Added by BALIK_10

Shortcut	Action
Ctrl+B	Open BibTeX export (frmBibTexExport). Scope follows current filter.

BALIK_11 — Additional Patch

BALIK_11.bas extends frmDetailTaxa with a free-text NOTES field, replaces the native Access tab strip with a left-side navigation sidebar, and fixes CARDINAL_PROCESSES display after clearing the search filter. Install after BALIK_10 using the standard procedure (File → Import File, then InstallBALIK11 in the Immediate Window, followed by CreateDetailFormTaxa_Ultimate).

What BALIK_11 Changes

Change	Description
NOTES field in TAXONOMY tab	A new NOTES field is added at the bottom of the TAXONOMY tab. It is stored in the NOTES column of Taxa_Poznamky (added automatically by ALTER TABLE). Height is 2400 twips. The field is unbound and uses LoadNotesField() / SaveNotesField() for DAO read/write. A small blue [S] button beside the field saves on demand. NOTES is also saved automatically when navigating to another record or closing the form.
Left navigation sidebar	The native TabControl tab strip is hidden (tc.Style = 2). Five dark buttons (1800 twips wide) are placed in a vertical column on the left edge: TAXONOMY, DESCRIPTION, OCCURRENCE, MATERIAL, REL-SHIPS. Active section highlighted in blue (#005580); inactive buttons dark grey (#2B2B2B). Clicking any button calls NavToSection(idx). GRID_START_X is shifted to 2000 twips to keep fields clear of the sidebar.
CARDINAL_PROCESSES fix after filter clear	After clearing the search filter in frmDark, fields such as CARDINAL_PROCESSES appeared empty in frmDetailTaxa until manual navigation. Fix: RefreshDetailIfOpen() is added to frmDark and called at the end of btnClear_Click. It calls df.Requery on frmDetailTaxa if open.

Installation Procedure

1. Open Hyolith_Database.accdb and press Alt+F11 to open the VBA Editor.
2. File → Import File → BALIK_11.bas.
3. Ctrl+G → Immediate Window → type: InstallBALIK11 → Enter. Confirm message box.
4. Immediate Window → type: CreateDetailFormTaxa_Ultimate → Enter. This regenerates frmDetailTaxa with all BALIK_11 changes applied.
5. Save the database (Ctrl+S in Access).